Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Live Oak Dairy E

Physical address of dairy:

8450 S Hwy 33	Gustine	Merced	95322
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1920

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X030-X021-XXXX

B. OPERATORS

Nyman, Brad

Operator name: Nyman, Brad

Telephone no.: (209) 495-8821 (Landline) (Cellular)

P.O. Box 1331	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Nyman, Brad

Legal owner name: Nyman, Brad

Telephone no.: (209) 495-8821 (Landline) (Cellular)

P.O. Box 1331	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	375	375	150	0
Number under roof	1,000	200	0	0	0	0
Maximum number	1,000	200	375	375	150	0
Average number	1,000	200	375	375	150	0
Avg live weight (lbs)	1,400	1,450	900	650		

Predominant milk cow breed: Jersey

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 36,379.56 *tons per reporting period*

Total nitrogen from manure: 450,720.10 *lbs per reporting period*

After ammonia losses (30% loss applied): 315,504.07 *lbs per reporting period*

Total phosphorus from manure: 74,182.86 *lbs per reporting period*

Total potassium from manure: 191,509.23 *lbs per reporting period*

Total salt from manure: 516,840.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 16,152,000 *gallons*

Total nitrogen generated: 64,879.90 *lbs*

Total phosphorus generated: 8,553.28 *lbs*

Total potassium generated: 86,434.44 *lbs*

Total salt generated: 691,795.16 *lbs*

	16,152,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	16,152,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well	Ground water
CCID	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Husman 1	20	20	2	manure	X070-X050-X002-XXXX
Husman 2	136	136	1	none	X070-X050-X009-XXXX X070-X050-X011-XXXX
Husman 3	175	175	1	none	X070-X005-X012-XXXX X070-X050-X010-XXXX X070-X050-X011-XXXX
Husman 4	156	156	0	none	X070-X080-X001-XXXX
Husman 5	240	240	0	none	X070-X050-X013-XXXX X070-X050-X014-XXXX
Jensen 1	25	25	2	process wastewater	X070-X030-X021-XXXX
Jensen 2	25	25	2	process wastewater	X070-X030-X021-XXXX
Jensen 3	25	25	2	process wastewater	X070-X030-X021-XXXX
Jensen 4	30	30	2	process wastewater	X070-X030-X021-XXXX
Jensen 5	50	50	2	process wastewater	X070-X030-X021-XXXX
Jensen 6	38	38	2	process wastewater	X070-X030-X021-XXXX
Whitworth 1	45	45	2	manure	X069-X011-X006-XXXX
Whitworth 2	20	20	2	manure	X069-X011-X006-XXXX
Whitworth 3	30	30	2	manure	X069-X014-X008-XXXX
Totals for areas that were used for application	308	308	20		
Totals for areas that were not used for application	707	707	2		
Land application area totals	1,015	1,015	22		

B. CROPS AND HARVESTS

Husman 1

Field name: Husman 1

Husman 1

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	180.00 *ton*	As-is		65.9	5,370.00	1,070.00	8,630.00		9.62

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	10.00	100.00	16.00	83.00	0.00
Total actual harvest content	9.00	96.66	19.26	155.34	590.48

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	494.00 *ton*	As-is		70.0	2,850.00	520.00	3,320.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	24.70	140.79	25.69	164.01	844.74

Husman 2

Field name: Husman 2

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 136 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	370.00 *ton*	As-is		61.1	5,380.00	1,210.00	9,370.00		8.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	80.00	12.80	66.40	0.00
Total actual harvest content	2.72	29.27	6.58	50.98	179.28

Husman 3

Field name: Husman 3

11/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 175 Plant date: 11/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/26/2023	610.00 *ton*	As-is		63.8	5,590.00	1,150.00	9,300.00		8.39

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	80.00	12.80	66.40	0.00
Total actual harvest content	3.49	38.97	8.02	64.83	211.73

Jensen 1

Field name: Jensen 1

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	256.00 *ton*	As-is		53.6	7,450.00	1,420.00	12,400.00		8.99

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.24	152.58	29.08	253.95	854.29

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	485.00 *ton*	As-is		70.0	3,510.00	660.00	3,440.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	19.40	136.19	25.61	133.47	663.48

Jensen 2

Field name: Jensen 2

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	395.00 *ton*	As-is		70.8	4,350.00	820.00	6,210.00		9.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.80	137.46	25.91	196.24	906.11

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	515.00 *ton*	As-is		70.0	3,240.00	700.00	3,380.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	20.60	133.49	28.84	139.26	729.24

Jensen 3

Field name: Jensen 3

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	359.00 *ton*	As-is		68.8	4,650.00	950.00	7,910.00		9.83

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.36	133.55	27.28	227.18	880.83

Jensen 3

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	505.00 *ton*	As-is		70.0	3,300.00	690.00	3,310.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	20.20	133.32	27.88	133.72	606.00

Jensen 4

Field name: Jensen 4

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	435.00 *ton*	As-is		63.6	5,240.00	1,020.00	7,890.00		8.17

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.50	151.96	29.58	228.81	862.43

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	625.00 *ton*	As-is		70.0	3,180.00	660.00	3,130.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	20.83	132.50	27.50	130.42	550.00

Jensen 5

Field name: Jensen 5

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	490.00 *ton*	As-is		55.4	8,730.00	1,210.00	12,500.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.80	171.11	23.72	245.00	961.58

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	910.00 *ton*	As-is		70.0	3,390.00	740.00	3,370.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	18.20	123.40	26.94	122.67	589.68

Jensen 6

Field name: Jensen 6

11/26/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/30/2023	485.00 *ton*	As-is		61.2	5,600.00	1,270.00	13,300.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.76	142.95	32.42	339.50	1,109.27

Jensen 6

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/25/2023	785.00 *ton*	As-is		70.0	3,120.00	690.00	3,360.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	20.66	128.91	28.51	138.82	743.68

Whitworth 1

Field name: Whitworth 1

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 45 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/31/2023	407.00 *ton*	As-is		59.8	7,850.00	1,250.00	12,600.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.04	142.00	22.61	227.92	821.71

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 45 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	705.00 *ton*	As-is		70.0	4,080.00	750.00	3,430.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	15.67	127.84	23.50	107.47	582.80

Whitworth 2

Field name: Whitworth 2

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/31/2023	185.00 *ton*	As-is		63.4	7,320.00	1,110.00	11,300.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.25	135.42	20.54	209.05	758.35

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	364.00 *ton*	As-is		70.0	3,420.00	610.00	3,190.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	18.20	124.49	22.20	116.12	556.92

Whitworth 3

Field name: Whitworth 3

11/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/31/2023	288.00 *ton*	As-is		63.8	7,140.00	1,130.00	11,300.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.60	137.09	21.70	216.96	834.05

Whitworth 3

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	512.00 *ton*	As-is		70.0	3,330.00	730.00	3,290.00		5.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	17.07	113.66	24.92	112.30	512.00

NUTRIENT BUDGET

A. LAND APPLICATIONS

Husman 1 - 11/13/2022: Oats, silage-soft dough

Field name: Husman 1

Crop: Oats, silage-soft dough — Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	133.64	58.40	319.04	0.00	125.00 *ton*
Application event totals		133.64	58.40	319.04	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Husman 1 - 06/29/2023: Corn, silage

Field name: Husman 1

Crop: Corn, silage — Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	152.51	65.36	445.09	4,139.63	250.00 *ton*
Application event totals		152.51	65.36	445.09	4,139.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	227.00	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	227.00	

Husman 1 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	174.62	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	174.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	157.16	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	157.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	139.70	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	157.16	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	157.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	139.70	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Husman 2 - 11/13/2022: Oats, silage-soft dough

Husman 2 - 11/13/2022: Oats, silage-soft dough

Field name: Husman 2

Crop: Oats, silage-soft dough

Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	1.28	135,000.00 *gal*
Application event totals		0.00	0.00	0.00	1.28	

Husman 3 - 11/13/2022: Oats, silage-soft dough

Field name: Husman 3

Crop: Oats, silage-soft dough

Plant date: 11/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	1.00	135,000.00 *gal*
Application event totals		0.00	0.00	0.00	1.00	

Jensen 1 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 1

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	66.81	10.17	108.15	748.25	240,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	132.71	2,565,000.00 *gal*
Application event totals		66.81	10.17	108.15	880.96	

Jensen 1 - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.26	8.77	46.14	549.89	576,000.00 *gal*
Ag Well	Ground water	11.88	0.00	0.00	431.52	1,944,000.00 *gal*
Application event totals		65.13	8.77	46.14	981.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.94	6.58	34.61	412.42	432,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	125.73	2,430,000.00 *gal*
Application event totals		39.94	6.58	34.61	538.14	

Jensen 1 - 06/30/2023: Corn, silage

Field name: Jensen 1

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	6.31	92.87	728.70	288,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	167.63	3,240,000.00 *gal*
Application event totals		65.66	6.31	92.87	896.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	195.57	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.57	

Jensen 1 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	6.31	92.87	728.70	288,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	153.66	2,970,000.00 *gal*
Application event totals		65.66	6.31	92.87	882.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	188.59	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	188.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	50.57	7.04	85.08	467.21	288,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	146.68	2,835,000.00 *gal*
Application event totals		50.57	7.04	85.08	613.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	181.60	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	181.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	195.57	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.57	

Jensen 2 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 2

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Jensen 2 - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.13	9.16	97.34	673.42	216,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	139.70	2,700,000.00 *gal*
Application event totals		60.13	9.16	97.34	813.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well	Ground water	11.88	0.00	0.00	431.52	1,944,000.00 *gal*
Application event totals		11.88	0.00	0.00	431.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.26	8.77	46.14	549.89	576,000.00 *gal*
Application event totals		53.26	8.77	46.14	549.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.94	6.58	34.61	412.42	432,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	125.73	2,430,000.00 *gal*
Application event totals		39.94	6.58	34.61	538.14	

Jensen 2 - 06/30/2023: Corn, silage

Field name: Jensen 2

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Jensen 2 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	6.31	92.87	728.70	288,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	181.60	3,510,000.00 *gal*
Application event totals		65.66	6.31	92.87	910.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	195.57	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	6.31	92.87	728.70	288,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	146.68	2,835,000.00 *gal*
Application event totals		65.66	6.31	92.87	875.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	181.60	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	181.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.14	5.86	70.90	389.34	240,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	146.68	2,835,000.00 *gal*
Application event totals		42.14	5.86	70.90	536.02	

Jensen 2 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	188.59	3,645,000.00 *gal*
Application event totals		0.00	0.00	0.00	188.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	181.60	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	181.60	

Jensen 3 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 3

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.26	8.77	46.14	549.89	576,000.00 *gal*
Ag Well	Ground water	11.88	0.00	0.00	431.52	1,944,000.00 *gal*
Application event totals		65.13	8.77	46.14	981.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.94	6.58	34.61	412.42	432,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	125.73	2,430,000.00 *gal*
Application event totals		39.94	6.58	34.61	538.14	

Jensen 3 - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.49	11.19	118.97	823.07	264,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	160.65	3,105,000.00 *gal*
Application event totals		73.49	11.19	118.97	983.72	

Jensen 3 - 06/30/2023: Corn, silage

Field name: Jensen 3

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	6.31	92.87	728.70	288,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	167.63	3,240,000.00 *gal*
Application event totals		65.66	6.31	92.87	896.33	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	195.57	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.57	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	6.31	92.87	728.70	288,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	153.66	2,970,000.00 *gal*
Application event totals		65.66	6.31	92.87	882.36	

Jensen 3 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	195.57	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.14	5.86	70.90	389.34	240,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	153.66	2,970,000.00 *gal*
Application event totals		42.14	5.86	70.90	543.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	195.57	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	195.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	174.62	3,375,000.00 *gal*
Application event totals		0.00	0.00	0.00	174.62	

Jensen 4 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 4

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Jensen 4 - 11/26/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.95	11.87	126.18	872.95	336,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	139.70	3,240,000.00 *gal*
Application event totals		77.95	11.87	126.18	1,012.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.38	7.31	38.45	458.24	576,000.00 *gal*
Ag Well	Ground water	9.90	0.00	0.00	359.60	1,944,000.00 *gal*
Application event totals		54.28	7.31	38.45	817.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.38	7.31	38.45	458.24	576,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	139.70	3,240,000.00 *gal*
Application event totals		44.38	7.31	38.45	597.94	

Jensen 4 - 06/30/2023: Corn, silage

Field name: Jensen 4

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.96	7.01	103.18	809.67	384,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	1,862.60	43,200,000.00 *gal*
Application event totals		72.96	7.01	103.18	2,672.27	

Jensen 4 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.96	7.01	103.18	809.67	384,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	151.34	3,510,000.00 *gal*
Application event totals		72.96	7.01	103.18	961.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	197.90	4,590,000.00 *gal*
Application event totals		0.00	0.00	0.00	197.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.14	5.86	70.90	389.34	288,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	139.70	3,240,000.00 *gal*
Application event totals		42.14	5.86	70.90	529.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	197.90	4,590,000.00 *gal*
Application event totals		0.00	0.00	0.00	197.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	180.44	4,185,000.00 *gal*
Application event totals		0.00	0.00	0.00	180.44	

Jensen 5 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 5

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.13	9.16	97.34	673.42	432,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	146.68	5,670,000.00 *gal*
Application event totals		60.13	9.16	97.34	820.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.26	8.77	46.14	549.89	1,152,000.00 *gal*
Ag Well	Ground water	11.88	0.00	0.00	431.52	3,888,000.00 *gal*
Application event totals		65.13	8.77	46.14	981.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.60	7.67	40.38	481.15	1,008,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	146.68	5,670,000.00 *gal*
Application event totals		46.60	7.67	40.38	627.83	

Jensen 5 - 06/30/2023: Corn, silage

Field name: Jensen 5

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	6.31	92.87	728.70	576,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	167.63	6,480,000.00 *gal*
Application event totals		65.66	6.31	92.87	896.33	

Jensen 5 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	8,100,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	65.66	6.31	92.87	728.70	576,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	153.66	5,940,000.00 *gal*
Application event totals		65.66	6.31	92.87	882.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	188.59	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	188.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.93	5.28	63.81	350.41	432,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	150.17	5,805,000.00 *gal*
Application event totals		37.93	5.28	63.81	500.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	132.71	5,130,000.00 *gal*
Application event totals		0.00	0.00	0.00	132.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	188.59	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	188.59	

Jensen 6 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 6

Crop: Oats, silage-soft dough

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.14	8.70	92.50	639.95	312,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	147.05	4,320,000.00 *gal*
Application event totals		57.14	8.70	92.50	786.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.56	8.65	45.54	542.65	864,000.00 *gal*
Ag Well	Ground water	11.72	0.00	0.00	425.85	2,916,000.00 *gal*
Application event totals		64.28	8.65	45.54	968.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.72	7.69	40.48	482.36	768,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	147.05	4,320,000.00 *gal*
Application event totals		46.72	7.69	40.48	629.41	

Jensen 6 - 06/30/2023: Corn, silage

Field name: Jensen 6

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.80	6.22	91.64	719.11	432,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	174.62	5,130,000.00 *gal*
Application event totals		64.80	6.22	91.64	893.73	

Jensen 6 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	193.00	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	193.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.80	6.22	91.64	719.11	432,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	147.05	4,320,000.00 *gal*
Application event totals		64.80	6.22	91.64	866.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	220.57	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	220.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	44.36	6.17	74.63	409.84	384,000.00 *gal*
CCID	Ground water	0.00	0.00	0.00	142.45	4,185,000.00 *gal*
Application event totals		44.36	6.17	74.63	552.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	202.19	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	202.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	147.05	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	147.05	

Whitworth 1 - 11/20/2022: Oats, silage-soft dough

Field name: Whitworth 1

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/29/2021	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	186.26	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	186.26	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	166.31	72.68	397.03	0.00	350.00 *ton*
Application event totals		166.31	72.68	397.03	0.00	

Whitworth 1 - 06/29/2023: Corn, silage

Field name: Whitworth 1

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	217.30	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	217.30	

Whitworth 1 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	186.26	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	162.98	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	162.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	162.98	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	162.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	139.70	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	139.70	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	147.90	53.04	249.09	6,716.48	600.00 *ton*
Application event totals		147.90	53.04	249.09	6,716.48	

Whitworth 1 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	147.46	5,130,000.00 *gal*
Application event totals		0.00	0.00	0.00	147.46	

Whitworth 2 - 11/20/2022: Oats, silage-soft dough

Field name: Whitworth 2

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	160.37	70.08	382.85	0.00	150.00 *ton*
Application event totals		160.37	70.08	382.85	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	157.16	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	157.16	

Whitworth 2 - 06/29/2023: Corn, silage

Field name: Whitworth 2

Crop: Corn, silage

Plant date: 06/29/2023

Whitworth 2 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	157.16	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	157.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	157.16	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	157.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	139.70	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	152.51	65.36	445.09	4,139.63	250.00 *ton*
Application event totals		152.51	65.36	445.09	4,139.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Whitworth 2 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	174.62	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	174.62	

09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	157.16	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	157.16	

Whitworth 3 - 11/20/2022: Oats, silage-soft dough

Field name: Whitworth 3

Crop: Oats, silage-soft dough

Plant date: 11/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	178.19	77.87	425.39	0.00	250.00 *ton*
Application event totals		178.19	77.87	425.39	0.00	

10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	186.26	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Whitworth 3 - 06/29/2023: Corn, silage

Field name: Whitworth 3

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	209.54	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	209.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	186.26	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	186.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	162.98	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	162.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	162.98	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	162.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	162.98	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	162.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	139.70	3,240,000.00 *gal*
Application event totals		0.00	0.00	0.00	139.70	

Whitworth 3 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	152.51	65.36	445.09	4,139.63	375.00 *ton*
Application event totals		152.51	65.36	445.09	4,139.63	

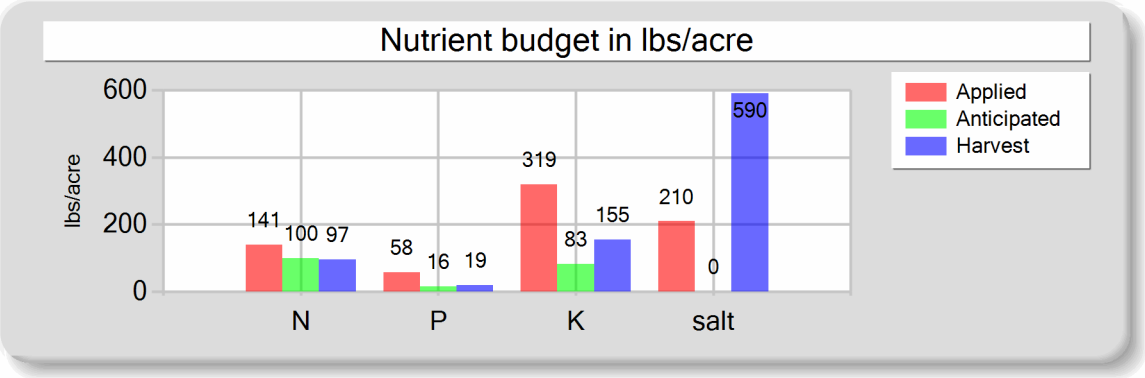
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	162.98	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	162.98	

B. NUTRIENT BUDGET

Husman 1 - 11/13/2022: Oats, silage-soft dough

Field name: Husman 1 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	133.64	58.40	319.04	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	209.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	140.64	58.40	319.04	209.54
Anticipated crop nutrient removal	100.00	16.00	83.00	0.00
Actual crop nutrient removal	96.66	19.26	155.34	590.48
Nutrient balance	43.98	39.14	163.70	-380.93
Applied to removed ratio	1.46	3.03	2.05	0.35

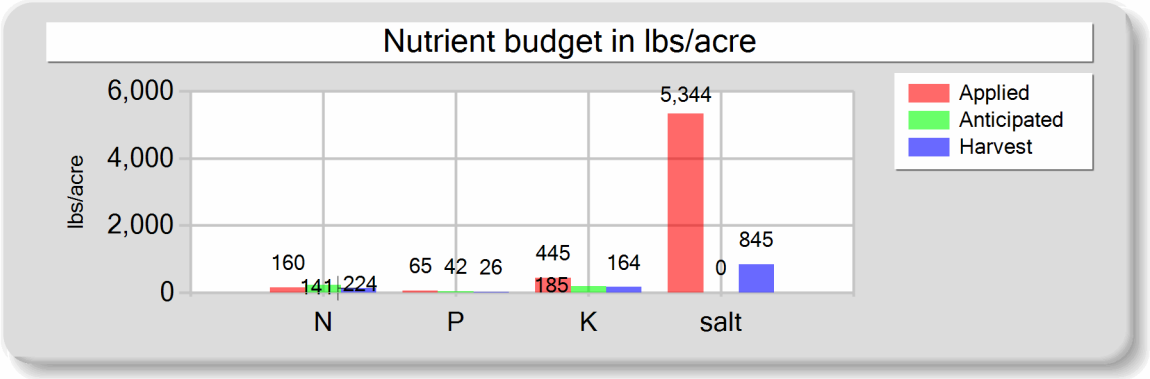
Fresh water applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
5.97	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Husman 1 - 06/29/2023: Corn, silage

Field name: Husman 1 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	152.51	65.36	445.09	4,139.63
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,204.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.51	65.36	445.09	5,344.50
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	140.79	25.69	164.01	844.74
Nutrient balance	18.72	39.67	281.08	4,499.76
Applied to removed ratio	1.13	2.54	2.71	6.33

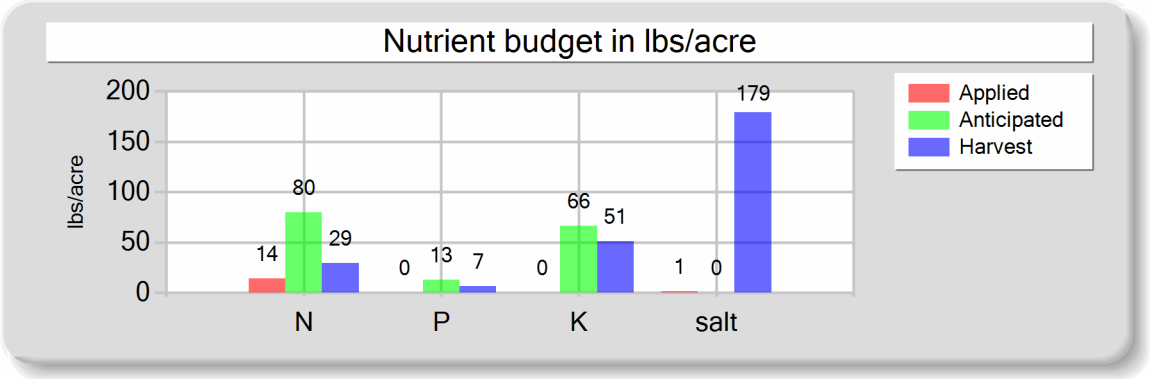
Fresh water applied	
18,630,000.00	*gallons*
686.08	*acre-inches*
34.30	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Husman 2 - 11/13/2022: Oats, silage-soft dough

Field name: Husman 2 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1.28
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	1.28
Anticipated crop nutrient removal	80.00	12.80	66.40	0.00
Actual crop nutrient removal	29.27	6.58	50.98	179.28
Nutrient balance	-15.27	-6.58	-50.98	-177.99
Applied to removed ratio	0.48	0.00	0.00	0.01

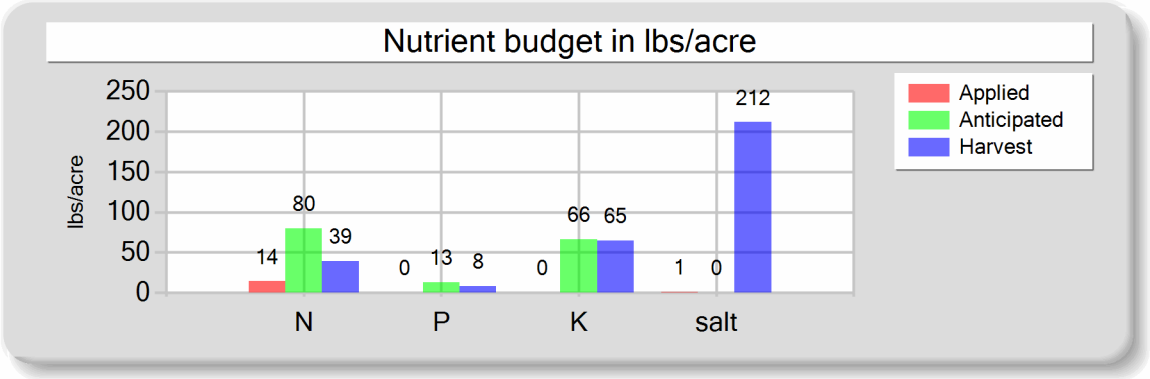
Fresh water applied	
135,000.00	*gallons*
4.97	*acre-inches*
0.04	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Husman 3 - 11/13/2022: Oats, silage-soft dough

Field name: Husman 3 Crop: Oats, silage-soft dough Plant date: 11/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	14.00	0.00	0.00	1.00
Anticipated crop nutrient removal	80.00	12.80	66.40	0.00
Actual crop nutrient removal	38.97	8.02	64.83	211.73
Nutrient balance	-24.97	-8.02	-64.83	-210.74
Applied to removed ratio	0.36	0.00	0.00	0.00

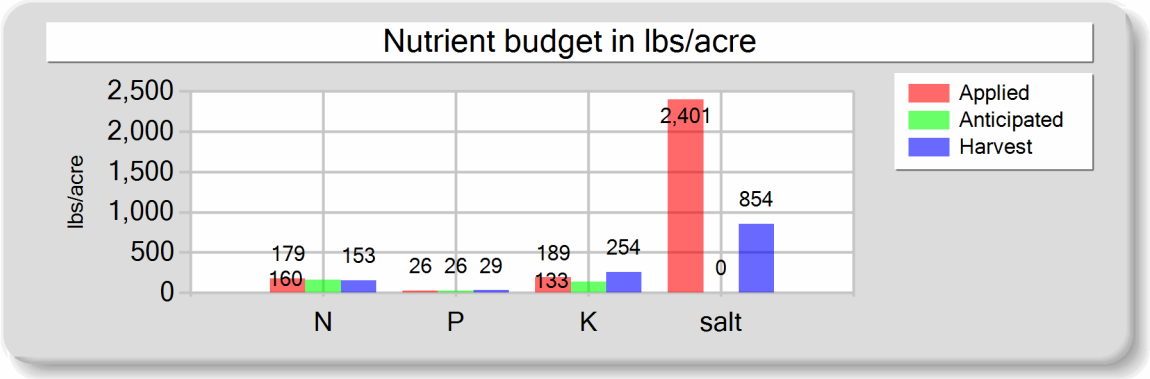
Fresh water applied	
135,000.00	*gallons*
4.97	*acre-inches*
0.03	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 1 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 1 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	160.02	25.52	188.90	1,710.55
Fresh water	11.88	0.00	0.00	689.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.89	25.52	188.90	2,400.51
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.58	29.08	253.95	854.29
Nutrient balance	26.31	-3.56	-65.05	1,546.22
Applied to removed ratio	1.17	0.88	0.74	2.81

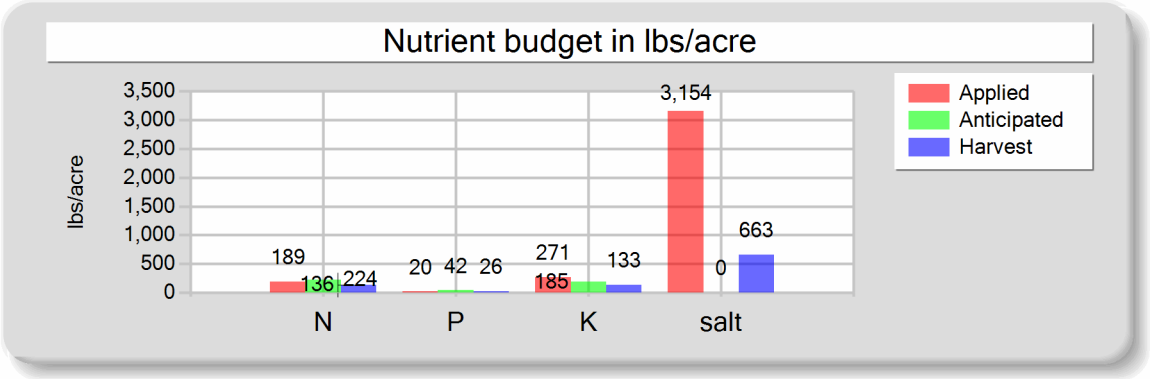
Fresh water applied	
6,939,000.00	*gallons*
255.54	*acre-inches*
10.22	*inches/acre*

Process wastewater applied	
1,248,000.00	*gallons*
45.96	*acre-inches*
1.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 1 - 06/30/2023: Corn, silage

Field name: Jensen 1 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	181.89	19.65	270.81	1,924.61
Fresh water	0.00	0.00	0.00	1,229.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	188.89	19.65	270.81	3,153.93
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	136.19	25.61	133.47	663.48
Nutrient balance	52.70	-5.96	137.34	2,490.45
Applied to removed ratio	1.39	0.77	2.03	4.75

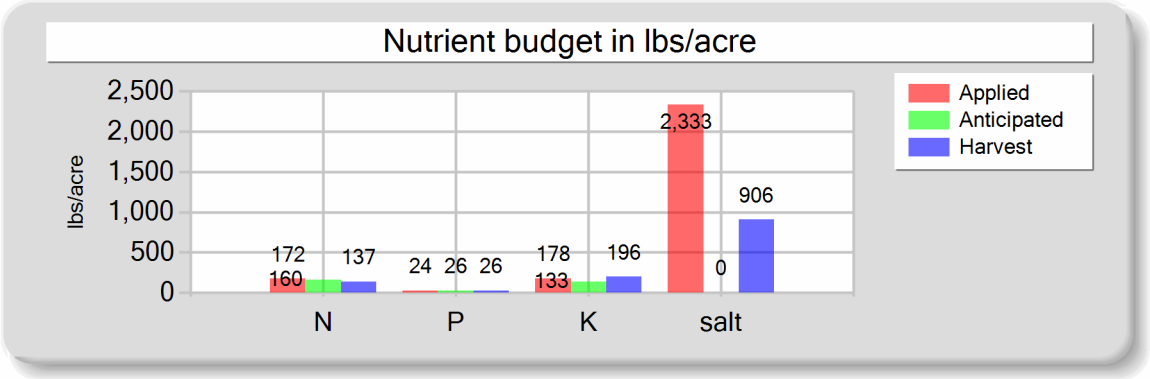
Fresh water applied	
23,760,000.00	*gallons*
875.00	*acre-inches*
35.00	*inches/acre*

Process wastewater applied	
864,000.00	*gallons*
31.82	*acre-inches*
1.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 2 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 2 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	153.33	24.50	178.09	1,635.73
Fresh water	11.88	0.00	0.00	696.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.21	24.50	178.09	2,332.67
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	137.46	25.91	196.24	906.11
Nutrient balance	34.75	-1.41	-18.15	1,426.56
Applied to removed ratio	1.25	0.95	0.91	2.57

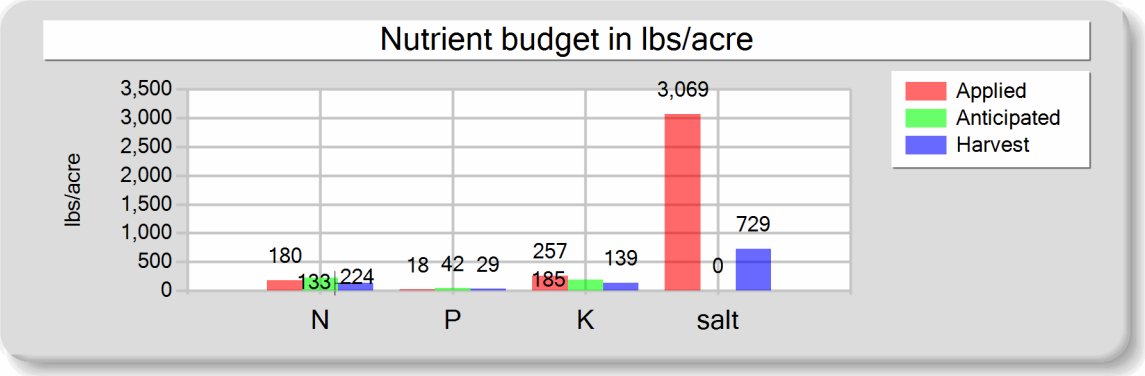
Fresh water applied	
7,074,000.00	*gallons*
260.51	*acre-inches*
10.42	*inches/acre*

Process wastewater applied	
1,224,000.00	*gallons*
45.08	*acre-inches*
1.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 2 - 06/30/2023: Corn, silage

Field name: Jensen 2 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.46	18.48	256.63	1,846.74
Fresh water	0.00	0.00	0.00	1,222.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.46	18.48	256.63	3,069.08
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	133.49	28.84	139.26	729.24
Nutrient balance	46.97	-10.36	117.37	2,339.84
Applied to removed ratio	1.35	0.64	1.84	4.21

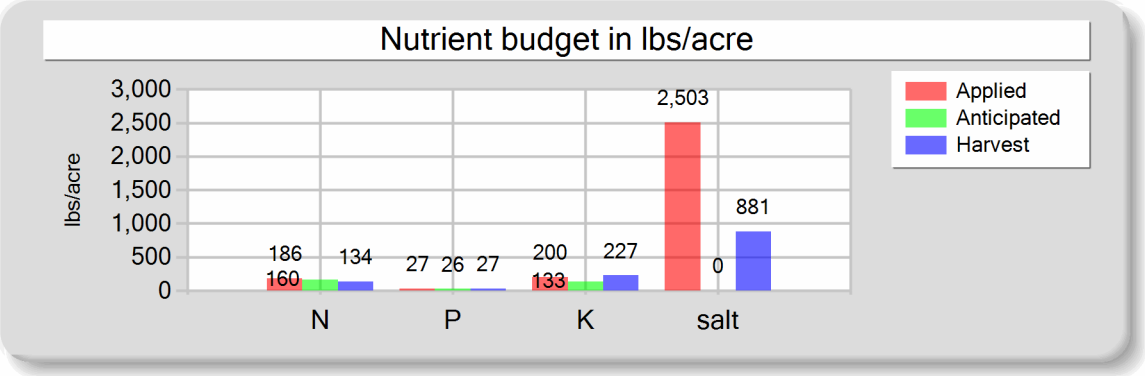
Fresh water applied	
23,625,000.00	*gallons*
870.03	*acre-inches*
34.80	*inches/acre*

Process wastewater applied	
816,000.00	*gallons*
30.05	*acre-inches*
1.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 3 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 3 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	166.70	26.53	199.72	1,785.38
Fresh water	11.88	0.00	0.00	717.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.57	26.53	199.72	2,503.27
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	133.55	27.28	227.18	880.83
Nutrient balance	52.02	-0.75	-27.46	1,622.44
Applied to removed ratio	1.39	0.97	0.88	2.84

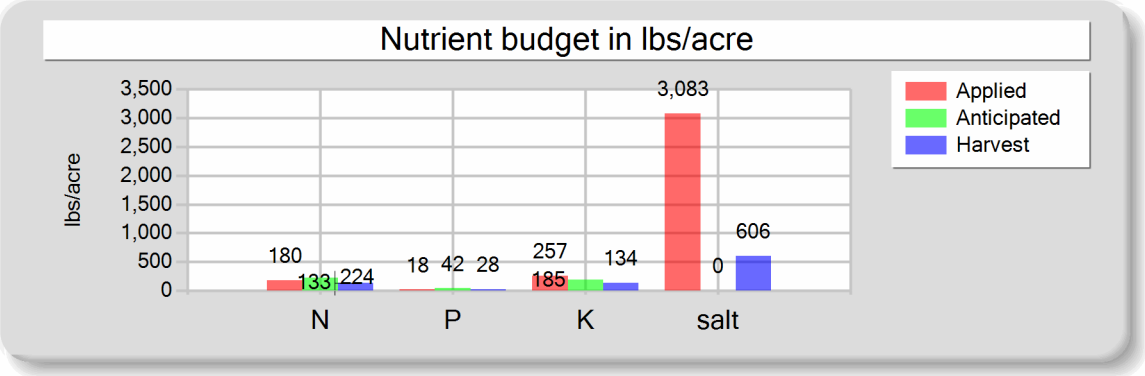
Fresh water applied	
7,479,000.00	*gallons*
275.43	*acre-inches*
11.02	*inches/acre*

Process wastewater applied	
1,272,000.00	*gallons*
46.84	*acre-inches*
1.87	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 3 - 06/30/2023: Corn, silage

Field name: Jensen 3 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.46	18.48	256.63	1,846.74
Fresh water	0.00	0.00	0.00	1,236.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.46	18.48	256.63	3,083.05
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	133.32	27.88	133.72	606.00
Nutrient balance	47.14	-9.40	122.91	2,477.05
Applied to removed ratio	1.35	0.66	1.92	5.09

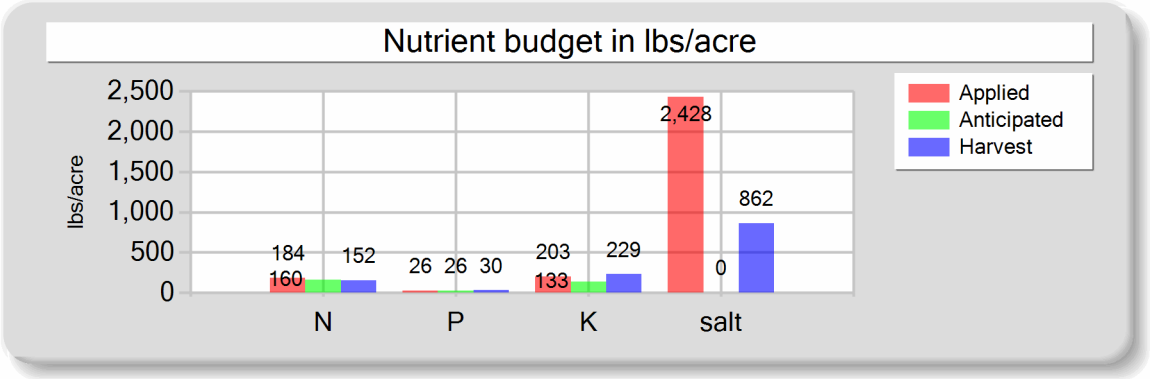
Fresh water applied	
23,895,000.00	*gallons*
879.97	*acre-inches*
35.20	*inches/acre*

Process wastewater applied	
816,000.00	*gallons*
30.05	*acre-inches*
1.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 4 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 4 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	166.71	26.48	203.08	1,789.44
Fresh water	9.90	0.00	0.00	638.99
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.61	26.48	203.08	2,428.43
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	151.96	29.58	228.81	862.43
Nutrient balance	31.65	-3.10	-25.73	1,566.00
Applied to removed ratio	1.21	0.90	0.89	2.82

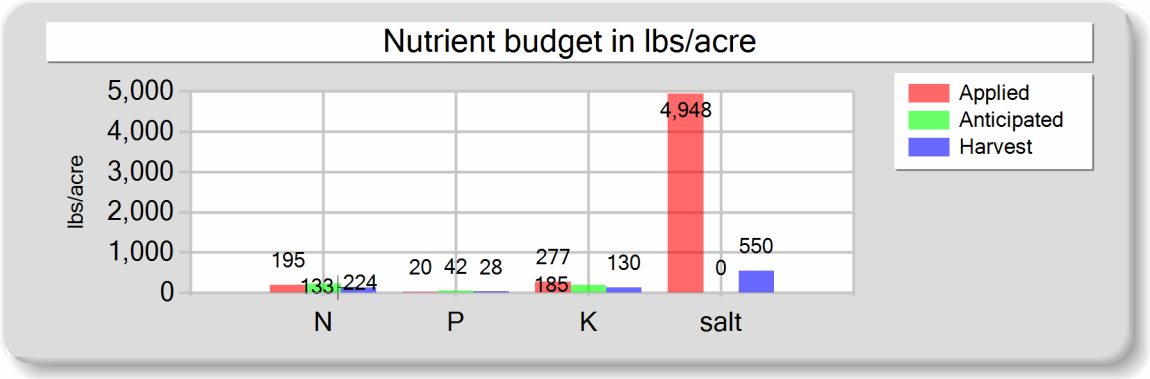
Fresh water applied	
8,424,000.00	*gallons*
310.23	*acre-inches*
10.34	*inches/acre*

Process wastewater applied	
1,488,000.00	*gallons*
54.80	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 4 - 06/30/2023: Corn, silage

Field name: Jensen 4 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	188.05	19.88	277.27	2,008.67
Fresh water	0.00	0.00	0.00	2,939.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	195.05	19.88	277.27	4,948.10
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	132.50	27.50	130.42	550.00
Nutrient balance	62.55	-7.62	146.85	4,398.10
Applied to removed ratio	1.47	0.72	2.13	9.00

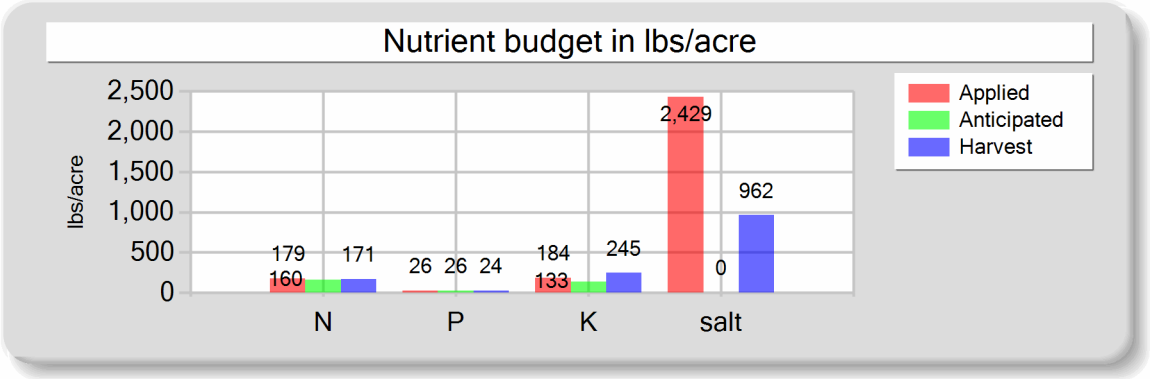
Fresh water applied	
68,175,000.00	*gallons*
2,510.65	*acre-inches*
83.69	*inches/acre*

Process wastewater applied	
1,056,000.00	*gallons*
38.89	*acre-inches*
1.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 5 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 5 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	159.99	25.60	183.86	1,704.46
Fresh water	11.88	0.00	0.00	724.88
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.87	25.60	183.86	2,429.35
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	171.11	23.72	245.00	961.58
Nutrient balance	7.76	1.88	-61.14	1,467.77
Applied to removed ratio	1.05	1.08	0.75	2.53

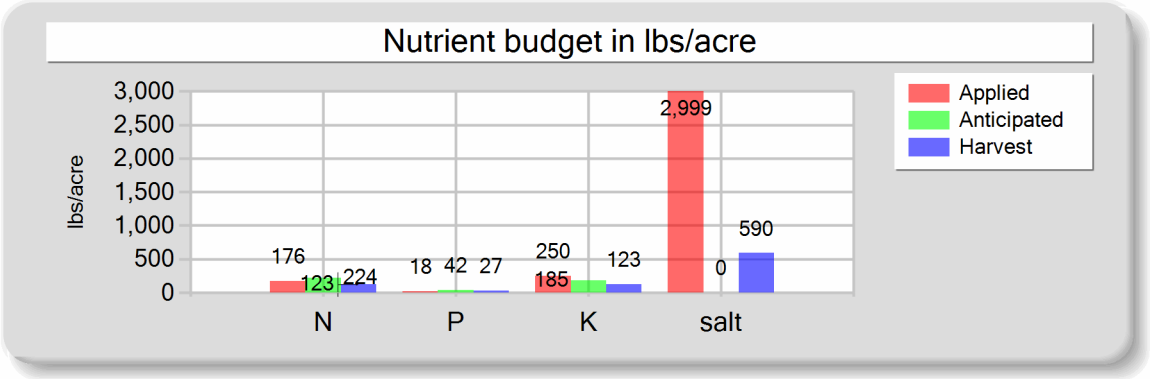
Fresh water applied	
15,228,000.00	*gallons*
560.80	*acre-inches*
11.22	*inches/acre*

Process wastewater applied	
2,592,000.00	*gallons*
95.45	*acre-inches*
1.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 5 - 06/30/2023: Corn, silage

Field name: Jensen 5 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	169.24	17.89	249.54	1,807.81
Fresh water	0.00	0.00	0.00	1,190.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	176.24	17.89	249.54	2,998.71
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	123.40	26.94	122.67	589.68
Nutrient balance	52.85	-9.05	126.87	2,409.03
Applied to removed ratio	1.43	0.66	2.03	5.09

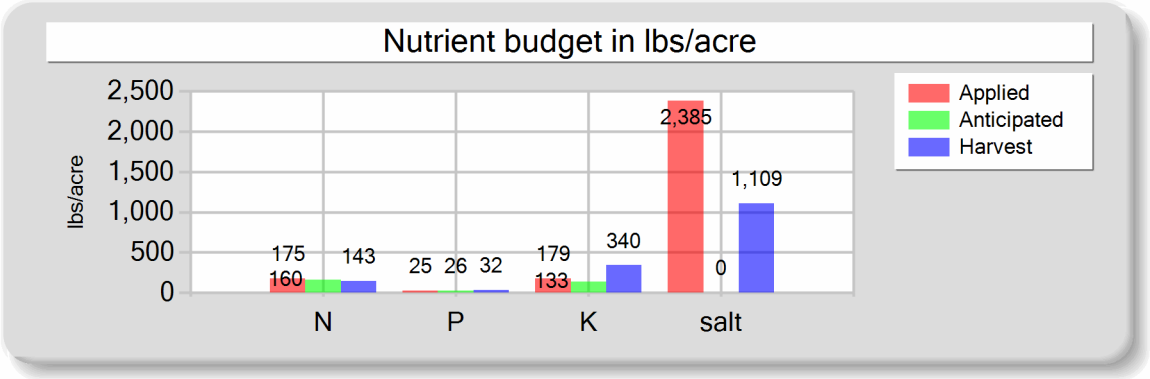
Fresh water applied	
46,035,000.00	*gallons*
1,695.31	*acre-inches*
33.91	*inches/acre*

Process wastewater applied	
1,584,000.00	*gallons*
58.33	*acre-inches*
1.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 6 - 11/26/2022: Oats, silage-soft dough

Field name: Jensen 6 Crop: Oats, silage-soft dough Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	156.42	25.04	178.51	1,664.96
Fresh water	11.72	0.00	0.00	719.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	175.14	25.04	178.51	2,384.90
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	142.95	32.42	339.50	1,109.27
Nutrient balance	32.19	-7.37	-160.99	1,275.63
Applied to removed ratio	1.23	0.77	0.53	2.15

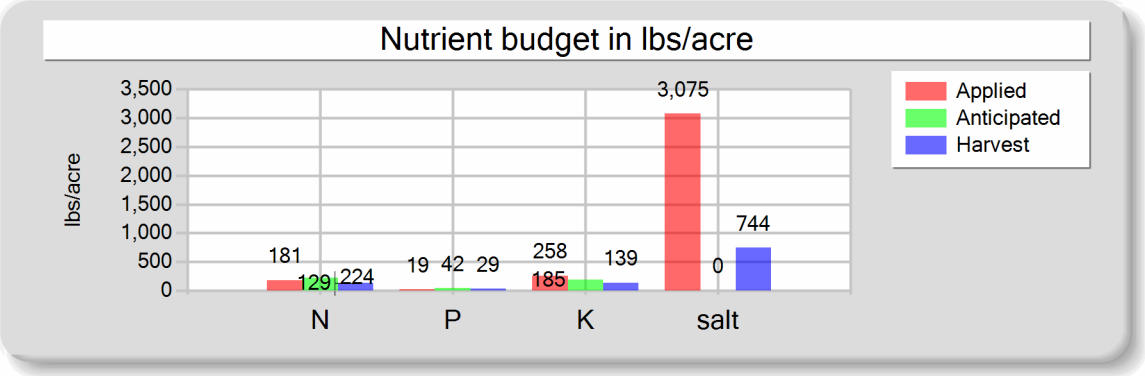
Fresh water applied	
11,556,000.00	*gallons*
425.57	*acre-inches*
11.20	*inches/acre*

Process wastewater applied	
1,944,000.00	*gallons*
71.59	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jensen 6 - 06/30/2023: Corn, silage

Field name: Jensen 6 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	173.95	18.62	257.92	1,848.06
Fresh water	0.00	0.00	0.00	1,226.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.95	18.62	257.92	3,074.99
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	128.91	28.51	138.82	743.68
Nutrient balance	52.04	-9.89	119.10	2,331.30
Applied to removed ratio	1.40	0.65	1.86	4.13

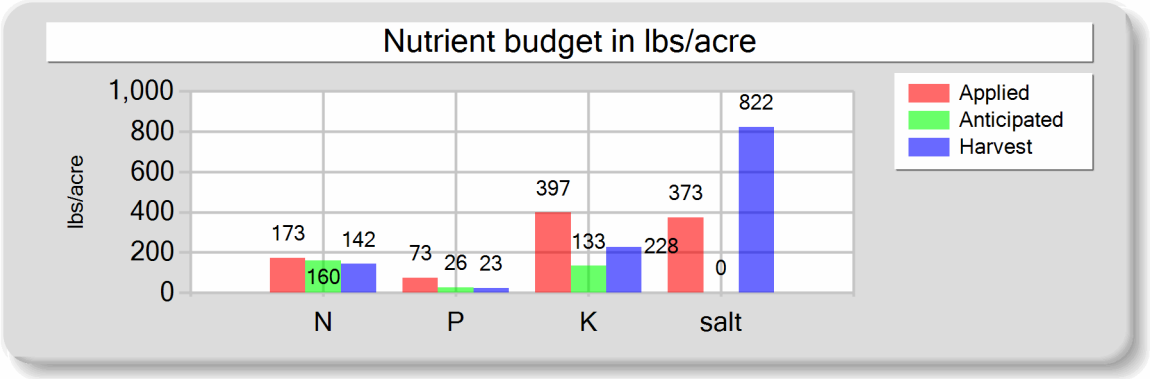
Fresh water applied	
36,045,000.00	*gallons*
1,327.41	*acre-inches*
34.93	*inches/acre*

Process wastewater applied	
1,248,000.00	*gallons*
45.96	*acre-inches*
1.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

Whitworth 1 - 11/20/2022: Oats, silage-soft dough

Field name: Whitworth 1 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	166.31	72.68	397.03	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	372.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.31	72.68	397.03	372.52
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	142.00	22.61	227.92	821.71
Nutrient balance	31.31	50.07	169.11	-449.19
Applied to removed ratio	1.22	3.21	1.74	0.45

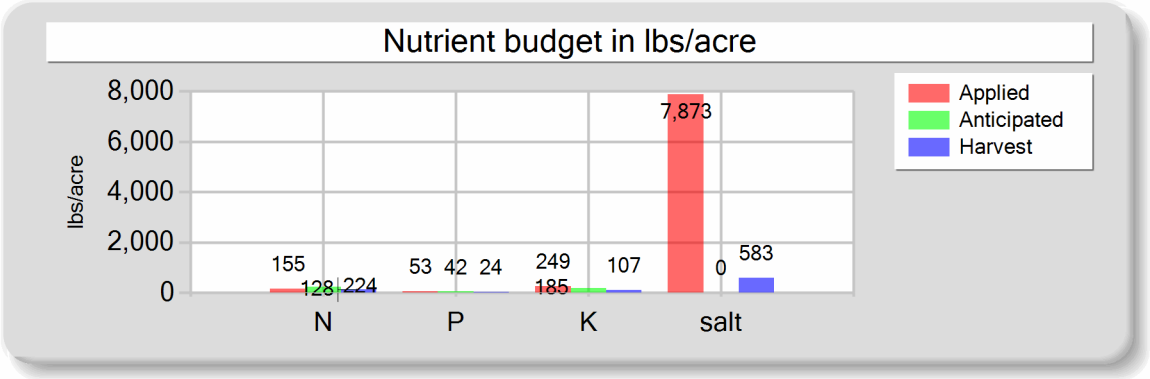
Fresh water applied	
12,960,000.00	*gallons*
477.27	*acre-inches*
10.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Whitworth 1 - 06/29/2023: Corn, silage

Field name: Whitworth 1 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	147.90	53.04	249.09	6,716.48
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,156.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.90	53.04	249.09	7,872.85
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	127.84	23.50	107.47	582.80
Nutrient balance	27.06	29.54	141.61	7,290.05
Applied to removed ratio	1.21	2.26	2.32	13.51

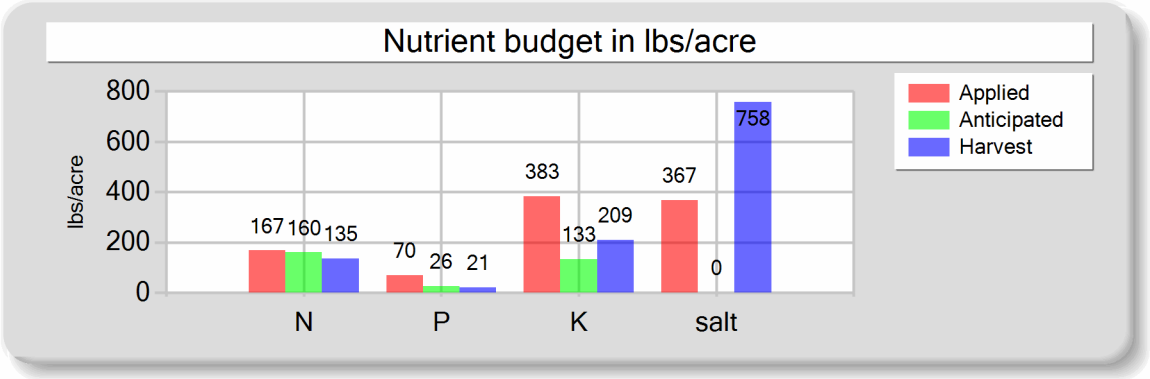
Fresh water applied	
40,230,000.00	*gallons*
1,481.53	*acre-inches*
32.92	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Whitworth 2 - 11/20/2022: Oats, silage-soft dough

Field name: Whitworth 2 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	160.37	70.08	382.85	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	366.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	167.37	70.08	382.85	366.70
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	135.42	20.54	209.05	758.35
Nutrient balance	31.95	49.55	173.80	-391.65
Applied to removed ratio	1.24	3.41	1.83	0.48

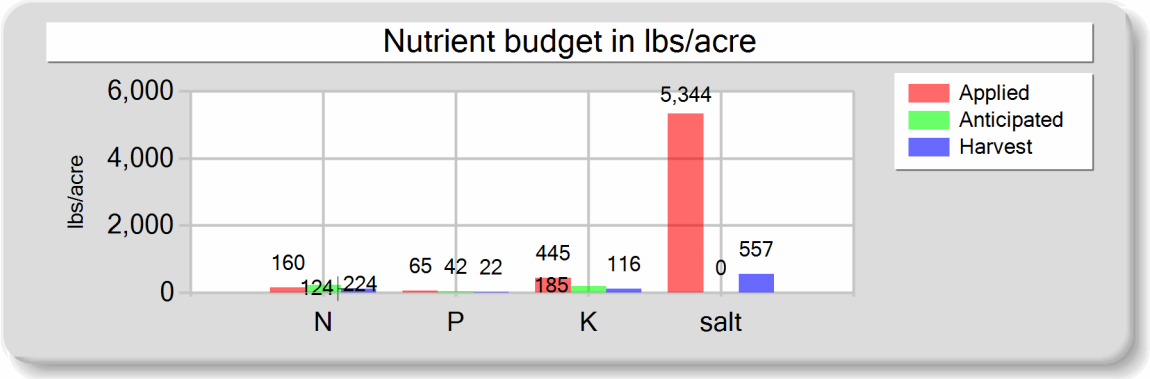
Fresh water applied	
5,670,000.00	*gallons*
208.81	*acre-inches*
10.44	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Whitworth 2 - 06/29/2023: Corn, silage

Field name: Whitworth 2 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	152.51	65.36	445.09	4,139.63
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,204.87
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.51	65.36	445.09	5,344.50
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	124.49	22.20	116.12	556.92
Nutrient balance	35.02	43.16	328.97	4,787.58
Applied to removed ratio	1.28	2.94	3.83	9.60

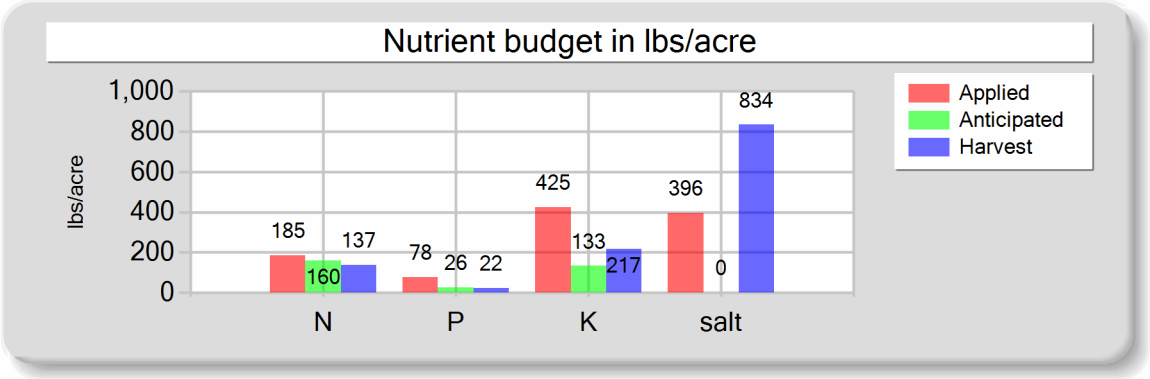
Fresh water applied	
18,630,000.00	*gallons*
686.08	*acre-inches*
34.30	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Whitworth 3 - 11/20/2022: Oats, silage-soft dough

Field name: Whitworth 3 Crop: Oats, silage-soft dough Plant date: 11/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.19	77.87	425.39	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	395.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.19	77.87	425.39	395.80
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	137.09	21.70	216.96	834.05
Nutrient balance	48.10	56.17	208.43	-438.24
Applied to removed ratio	1.35	3.59	1.96	0.47

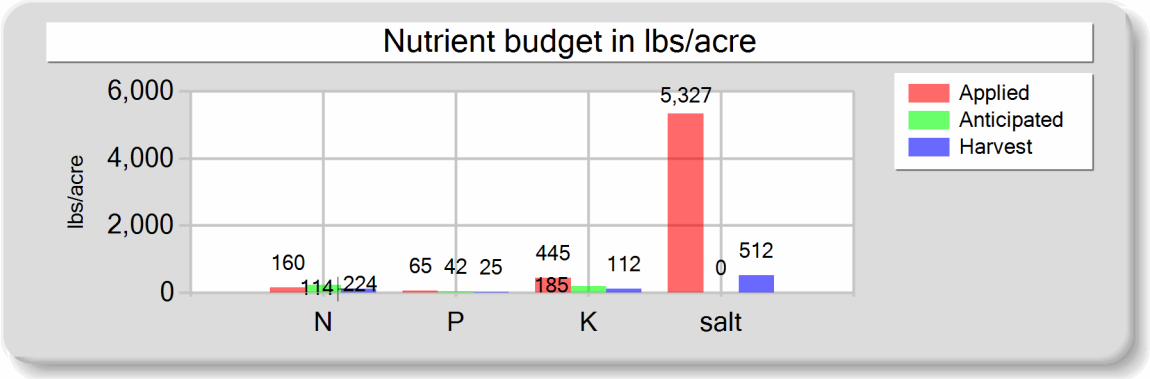
Fresh water applied	
9,180,000.00	*gallons*
338.07	*acre-inches*
11.27	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Whitworth 3 - 06/29/2023: Corn, silage

Field name: Whitworth 3 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	152.51	65.36	445.09	4,139.63
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,187.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.51	65.36	445.09	5,327.04
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	113.66	24.92	112.30	512.00
Nutrient balance	45.85	40.45	332.79	4,815.04
Applied to removed ratio	1.40	2.62	3.96	10.40

Fresh water applied	
27,540,000.00	*gallons*
1,014.20	*acre-inches*
33.81	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall SM

Sample and source description: Fall SM

Sample date: 10/21/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,300.00	7,560.00	41,300.00							0.00
DL	100.00	100.00	100.00							0.01

Spring SM

Sample and source description: Spring SM

Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	4,770.00	22,400.00							60.40
DL	100.00	100.00	100.00							0.01

Fall SM

Sample and source description: Fall SM

Sample date: 10/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 35.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,400.00	6,030.00	54,700.00							33.80
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 12/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	834.00	576.00	0.00	0.00	127.00	1,350.00								10.50	9,340
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 03/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	277.00	120.00	0.00	0.00	45.60	240.00								2,960.00	2,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 05/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	683.00	434.00	0.00	0.00	65.60	966.00								8,890.00	7,580
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 10/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	526.00	375.00	0.00	0.00	73.20	885.00								7,420.00	4,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Ag Well

Fresh Water

Sample description: Fresh Water

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	18.30	0.00	15.50								1,240.00	665
DL	0.50	0.50	0.50								1.00	10

CCID

Fresh Water

Sample description: Fresh Water

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								228.00	155
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

Husman 1

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			45.00				
DL			4.00				

Husman 2

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			23.00				
DL			4.00				

Husman 3

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			12.00				
DL			4.00				

Husman 4

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			7.00				
DL			4.00				

Husman 5

Husman 5

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			0.00				
DL			4.00				

Jensen 1

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			62.00				
DL			4.00				

Jensen 2

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			78.00				
DL			4.00				

Jensen 3

Jensen 3

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			67.00				
DL			4.00				

Jensen 4

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			67.00				
DL			4.00				

Jensen 5

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			69.00				
DL			4.00				

Jensen 6

Jensen 6

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00				
DL			4.00				

Whitworth 1

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			47.00				
DL			4.00				

Whitworth 2

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			67.00				
DL			4.00				

Whitworth 3

Whitworth 3

Soils

Sample and source description: Soils

Sample date: 11/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			57.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Husman 1 - 11/13/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,370.00	1,070.00	8,630.00		9.62
DL	100.00	100.00	100.00		0.01

Husman 1 - 06/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,850.00	520.00	3,320.00		5.70
DL	100.00	100.00	100.00		0.01

Husman 2 - 11/13/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,380.00	1,210.00	9,370.00		8.47
DL	100.00	100.00	100.00		0.01

Husman 3 - 11/13/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,590.00	1,150.00	9,300.00		8.39
DL	100.00	100.00	100.00		0.01

Jensen 1 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,450.00	1,420.00	12,400.00		8.99
DL	100.00	100.00	100.00		0.01

Jensen 1 - 06/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,510.00	660.00	3,440.00		5.70
DL	100.00	100.00	100.00		0.01

Jensen 2 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,350.00	820.00	6,210.00		9.82
DL	100.00	100.00	100.00		0.01

Jensen 2 - 06/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,240.00	700.00	3,380.00		5.90
DL	100.00	100.00	100.00		0.01

Jensen 3 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,650.00	950.00	7,910.00		9.83
DL	100.00	100.00	100.00		0.01

Jensen 3 - 06/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	690.00	3,310.00		5.00
DL	100.00	100.00	100.00		0.01

Jensen 4 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,240.00	1,020.00	7,890.00		8.17
DL	100.00	100.00	100.00		0.01

Jensen 4 - 06/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,180.00	660.00	3,130.00		4.40
DL	100.00	100.00	100.00		0.01

Jensen 5 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 55.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,730.00	1,210.00	12,500.00		11.00
DL	100.00	100.00	100.00		0.01

Jensen 5 - 06/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	740.00	3,370.00		5.40
DL	100.00	100.00	100.00		0.01

Jensen 6 - 11/26/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,270.00	13,300.00		11.20
DL	100.00	100.00	100.00		0.01

Jensen 6 - 06/30/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,120.00	690.00	3,360.00		6.00
DL	100.00	100.00	100.00		0.01

Whitworth 1 - 11/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,850.00	1,250.00	12,600.00		11.30
DL	100.00	100.00	100.00		0.01

Whitworth 1 - 06/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,080.00	750.00	3,430.00		6.20
DL	100.00	100.00	100.00		0.01

Whitworth 2 - 11/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,320.00	1,110.00	11,300.00		11.20
DL	100.00	100.00	100.00		0.01

Whitworth 2 - 06/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,420.00	610.00	3,190.00		5.10
DL	100.00	100.00	100.00		0.01

Whitworth 3 - 11/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 06/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,140.00	1,130.00	11,300.00		12.00
DL	100.00	100.00	100.00		0.01

Whitworth 3 - 06/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,330.00	730.00	3,290.00		5.00
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

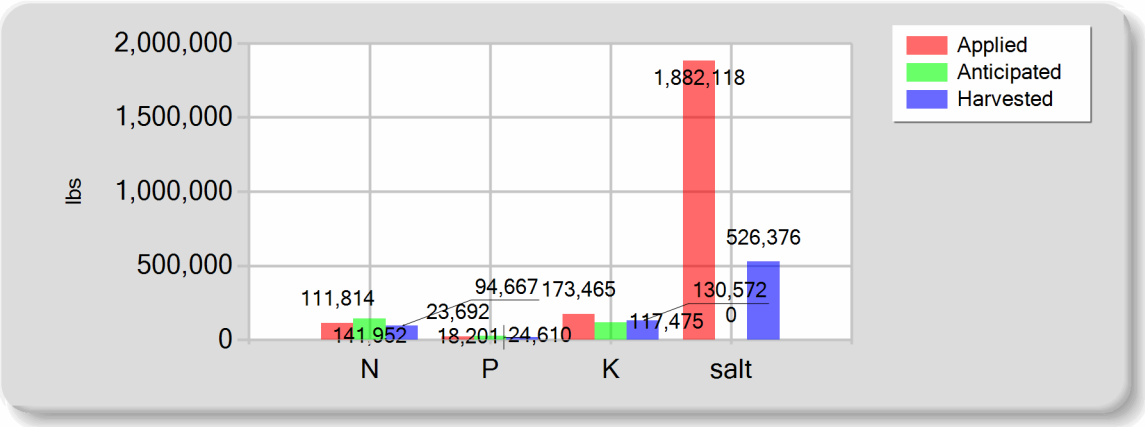
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

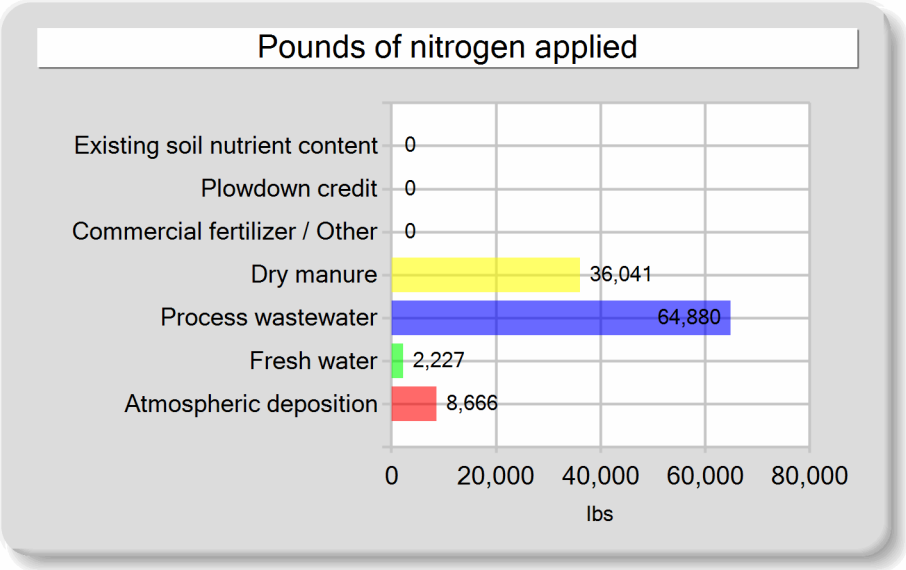
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

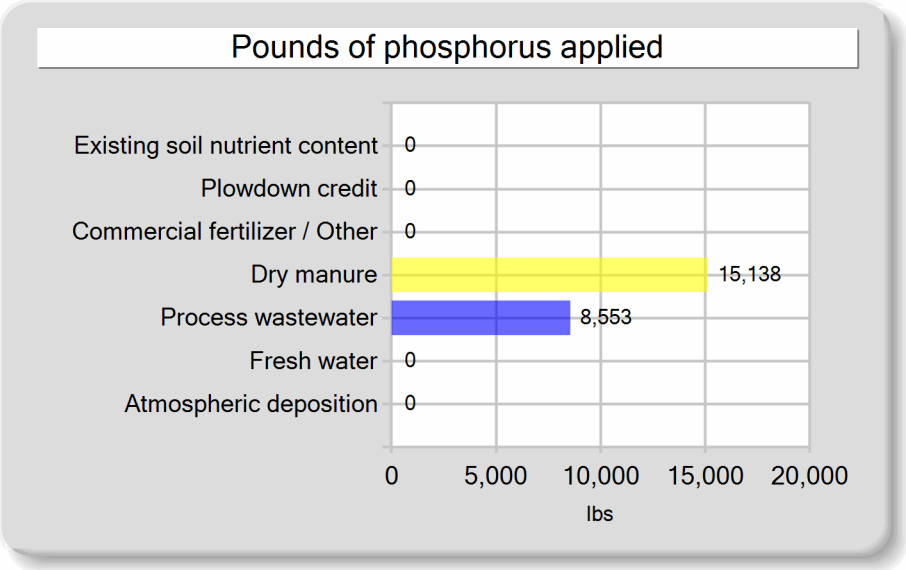
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	36,041.15	15,138.42	87,031.04	592,015.35
Process wastewater	64,879.90	8,553.28	86,434.44	691,795.16
Fresh water	2,226.56	0.00	0.00	598,307.08
Atmospheric deposition	8,666.00	0.00	0.00	0.00
Total nutrients applied	111,813.61	23,691.70	173,465.47	1,882,117.59
Anticipated crop nutrient removal	141,952.00	24,609.60	117,475.20	0.00
Actual crop nutrient removal	94,667.32	18,200.78	130,572.32	526,376.08
Nutrient balance	17,146.29	5,490.92	42,893.15	1,355,741.51
Applied to removed ratio	1.18	1.30	1.33	3.58

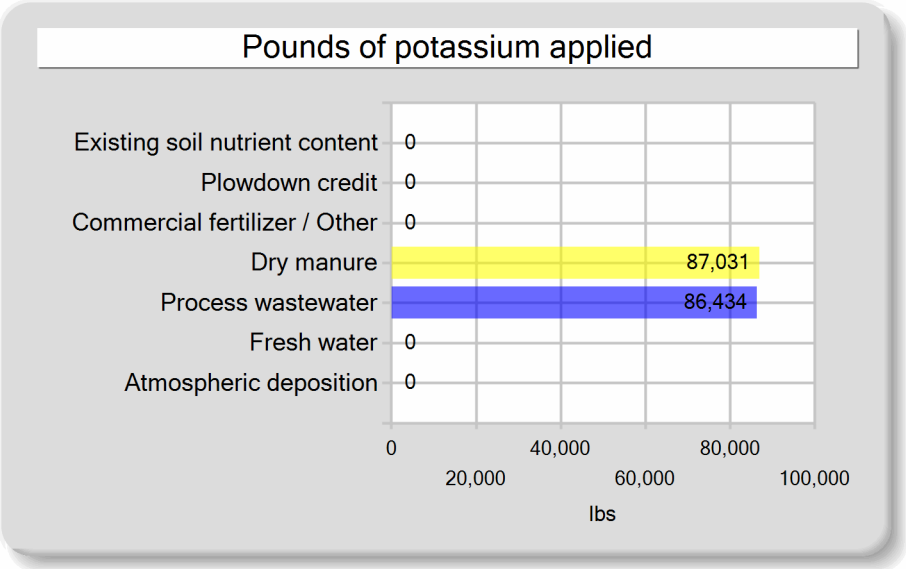
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

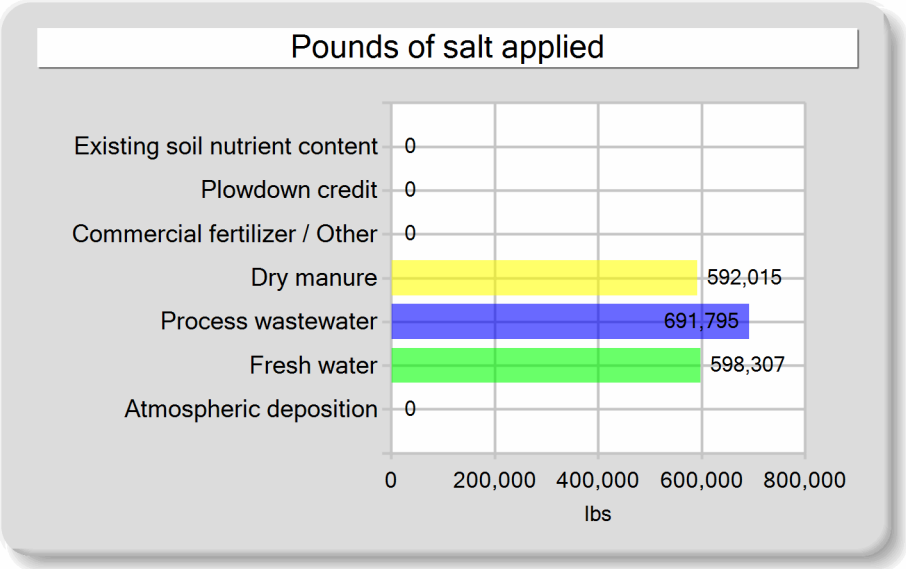


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Brad Nyman
Brad Nyman (Jun 19, 2024 09:28 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Brad Nyman	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 19, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Live Oak Dairy E
PO Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Barn	T3240030A	Electrical Conductivity (EC)	EPA 120.1 and	1.35	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	18.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	866	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.8 (T3240030A)	94.6 (T3240030A)	1.26
Electrical Conductivity (EC)	99.9			0.307
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Live Oak Dairy E
PO Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW-Yard	T3240030B	Electrical Conductivity (EC)	EPA 120.1 and	1.39	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	892	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.8 (T3240030A)	94.6 (T3240030A)	1.26
Electrical Conductivity (EC)	99.9			0.307
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)	1.65

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Live Oak Dairy E
PO Box 1331
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 8/28/23 12:36 PM
SUBMITTED BY: KJ
ANALYZED DATE: 8/29/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 14.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well	T3240030C	Electrical Conductivity (EC)	EPA 120.1 and	1.24	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	15.5	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	665	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	18.3	mg/L
		Soluble Salts (SALT-SOL)	Calculation	796	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	98.0	95.8 (T3240030A)	94.6 (T3240030A)			1.26
Electrical Conductivity (EC)	99.9			1.63 (T3240033B)	1.63 (Dup)	0.307
Nitrate Nitrogen (NO3-N)	99.9	101 (T3240027F)	99.6 (T3240027F)			1.65
Total Dissolved Solids (TDS)	104			905 (T3240033B)	910 (Dup)	0.551
Total Nitrogen (TN)	99.1			2.80 (T3240027F)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Live Oak E
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: FR
Sampled By: BN
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8/24	12:15	Domestic	DW-Barn		16.0°	✓				
2	↓	12:30	↓ well	DW-Yard		15.6°	✓				
3	✓	12:45	Irrigation well	Ag well		14.8°					H7
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st		FRAG				
2nd		DENELE ANALYTICAL, INC.	8-28-23 12:36			T3240030

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)

LOE 2023 Annual Report

Final Audit Report 2024-06-19

Created:	2024-06-19
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAACrPucjc2DUDhF-Ad71Y0zSOnNFK4J3dW

"LOE 2023 Annual Report" History

Document created by Joe Ramos (jramos@fragservices.com)
2024-06-19 - 3:20:31 PM GMT

Document emailed to nymanbrad@gmail.com for signature
2024-06-19 - 3:22:02 PM GMT

Email viewed by nymanbrad@gmail.com
2024-06-19 - 4:27:14 PM GMT

Signer nymanbrad@gmail.com entered name at signing as Brad Nyman
2024-06-19 - 4:28:00 PM GMT

Document e-signed by Brad Nyman (nymanbrad@gmail.com)
Signature Date: 2024-06-19 - 4:28:02 PM GMT - Time Source: server

Agreement completed.
2024-06-19 - 4:28:02 PM GMT